The Limits of the Expectations Hypothesis

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Abstract

Statistical tests reject the expectations hypothesis (EH) even at the very short end of the term structure, yet trading on the rejections earns little. Because these tests assume a linear model of expectations, it has remained open whether a less restrictive test would uncover economically meaningful departures. We test an implication of the EH that holds for any expectations process, namely that the excess return of rolling over short repos instead of holding a longer repo is unpredictable. The test is out of sample, on daily US general-collateral repo rates from overnight to one month over 1998–2026, with linear, tree-based, and neural-network forecasts estimated under one protocol. The in-sample Wald test rejects a true EH in 88% to 97% of simulated samples and is therefore uninformative. Out of sample, departures are predictable only at horizons of one and two weeks, a linear function of current rates captures them, and most of this predictability disappears when same-day composite quotes are excluded. Relaxing the functional form does not change the economic conclusion: trading on any of 22 forecasts earns at most 2.2 basis points a year before costs and 0.9 after observed bid–ask spreads, and even perfect foresight of the excess return would earn at most 4.1 basis points a year per unit of notional before costs. Statistical rejections of the EH at the short end thus coexist with a tight economic bound on its failure.

Keywords: Expectations hypothesis, Term premium, Repo rates, Forecast evaluation, Economic value, Machine learning
JEL: E43, G12, C53

1. Introduction

A cash investor can lock in a one-week repo rate today or roll over overnight repos for a week. Under the expectations hypothesis (EH) the choice does not matter up to a constant premium. Della Corte et al. (2008) (DST) reject the EH statistically for every pair of US repo maturities from overnight to three months, yet find that trading on the departures is worth at most 1.34 basis points a year before costs at a 1% volatility target and is not

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economically significant after costs. They caution that their tests rely on a linear vector autoregression (VAR) as the model of expectations. Every standard EH test is a joint test of the hypothesis and an expectations process (Guidolin and Thornton, 2018; Crump et al., 2024). A nonlinear process could therefore create spurious rejections or hide economically important departures.

This letter asks whether economically meaningful failure of the EH survives once the linear expectations model is removed. We test a restriction that holds for any expectations process: the excess return of rolling over holding is unpredictable with information available at the investment date. Its best forecast is a constant, estimated in real time by the prevailing mean. Any model that beats this forecast out of sample is evidence against the EH, and its forecasts can be traded as in DST. To relax the functional form, linear, regularised, tree-based, and neural-network forecasts compete under one protocol; the flexible models are a device, not a contribution in their own right.

Three results emerge. First, in-sample rejections are uninformative here: in samples simulated from a model in which the EH holds exactly, the Wald test of DST rejects in 88% to 97% of cases. Second, out of sample the EH fails only at horizons of one and two weeks, a linear function of current rates captures the predictability, and most of it is tied to same-day composite quotes. Third, the failure is economically bounded: no forecast earns more than 2.2 basis points a year before costs or 0.9 after observed bid–ask spreads, and even perfect foresight of the excess return would earn at most 4.1 basis points a year before costs.

The economic conclusion resembles DST's, but it now rests on weaker assumptions. DST evaluate in-sample forecasts of one linear model. Our evaluation is out of sample, does not depend on a model of expectations, extends to 2026, and bounds the value of any forecast. The letter thereby tests whether a classic finding is contingent on its method. It adds to out-of-sample evidence on the EH (Guidolin and Timmermann, 2009; Thornton and Valente, 2012; Sarno et al., 2016) and qualifies machine-learning evidence of large bond-premium predictability at long maturities (Bianchi et al., 2021): at the short end, flexible forecasts

do not change the economic verdict. Section 2 states the restriction, Section 3 the data and the forecast design, Section 4 the three results, and Section 5 concludes.

2. Testing the EH Without Modelling Expectations

Let $i_t^{(n)}$ be the rate at date t on a repo with a maturity of n business days, and let the short repo have maturity m, with $n = km$ for an integer k. Repo rates are simple ACT/360 rates. Holding the long repo returns $r_t^f = i_t^{(n)} D_t/360$ over its life of D_t calendar days. Rolling the short repo k times returns $r_t = \prod_{j=0}^{k-1}(1 + i_{t_j}^{(m)} d_j/360) - 1$, where d_j are the calendar days of the j-th roll. The annualised excess return is

$$xr_t^{(n,m)} = \left(r_t - r_t^f\right) 360/D_t. \tag{1}$$

With a constant term premium $\pi^{(n,m)}$ and rational expectations, the EH implies

$$E_t\left[xr_t^{(n,m)}\right] = -\pi^{(n,m)}. \tag{2}$$

The deviation $xr_t^{(n,m)} + \pi^{(n,m)}$ is then orthogonal to any function of the information at t, linear or not. The VAR test of DST imposes the same hypothesis through cross-equation restrictions that are valid only if expectations follow the VAR.

The restriction maintains three assumptions: the premium is constant, expectations are rational given the information at t, and quoted rates are prices at which investors can trade. Section 4.1 relaxes the first by allowing a slowly moving premium and examines the third with transaction-based rates. A rejection therefore means only that the excess return is predictable. Predictability may reflect a time-varying premium, expectation errors, or quote errors, and it says nothing about size. Economic failure requires that predictability be exploitable, which we measure separately. The next section describes how we take the restriction to the data.

3. Data and Forecast Design

We use Bloomberg composite closing quotes of US general-collateral Treasury repo rates for overnight (ON), one and two weeks (1W, 2W), and one month (1M), from 23 July 1998 to 30 September 2026. These are the four maturities whose quotes are updated throughout the sample; the 3W, 2M, and 3M composites have not changed since December 2015 while other repo rates rose above 5%, and we do not use them. The four maturities give the pairs ON-1W, ON-2W, ON-1M, and 1W-2W. Even these series contain frozen periods (Figure 1, Panel A), for instance ON and 1W in 2001–2004 and 1M from 2016 to 2019. We therefore set a quote to missing if it is unchanged for more than 20 business days, which removes between 18% (ON) and 42% (2W) of the observations. The results are similar with thresholds of 10 or 40 days and with a rule that uses only past quotes (Supplement, Section S6). Panel B of Figure 1 shows the resulting out-of-sample dates, which for every pair extend to 2026. US holidays are never used as forecast dates.

[Insert Figure 1 here]

On these data, every model forecasts the excess return of Equation (1) on each business day t from five daily lags of the pair's two rates, the information set of DST's VAR(5). Rolling windows contain 1,250 business days and include only excess returns realised by t. Models are re-estimated every 21 days and tuned yearly on the last 15% of each window, which is separated from the training targets by the overlap of the returns. Table S1 of the Supplement documents this common protocol. The EH forecast is the prevailing mean, and the VAR forecast is an ordinary least squares regression on the lags. The competing classes are regularised linear models, kernel and nearest-neighbour methods, tree ensembles, ridge regression on random features, and neural networks with one to five layers. All 22 forecasts are evaluated on the same dates.

Accuracy is the out-of-sample R^2_{oos} relative to the EH forecast (Campbell and Thompson, 2008). Individual forecasts are tested with Clark and West (2007), using Newey–West

variances with at least as many lags as the horizon. Because the EH concerns every forecast, the test of superior predictive ability (SPA) of Hansen (2005) asks whether any of the 22 forecasts beats the EH. Economic value is the annual performance fee (Fleming et al., 2001) that an investor with relative risk aversion of 5 would pay to switch from the EH strategy to a forecast-based strategy, as in DST. Positions are scaled to about one unit of notional. A bid–ask spread is charged when a position is renewed: either 4 basis points, as in DST, or the average of the observed mean spreads of the pair's two maturities, which range from 1 basis point (2W, 1M) to 6 (ON). To bound the value of any forecast, we also evaluate an investor with perfect foresight of the sign of the excess return.

4. Results

We report the evidence in the order of the three results: what survives out of sample, whether flexible functional forms change it, and what it is worth.

[Insert Table 1 here]

4.1. What survives out of sample?

Panel A of Table 1 shows why in-sample tests cannot settle the question. The Wald test of DST rejects the EH for all four pairs. Yet in samples simulated from a two-factor model in which the EH holds exactly, it also rejects in 88% to 97% of cases, in line with the size distortions of tests with near-integrated rates (Bekaert and Hodrick, 2001), and against these simulated distributions no rejection in the data is significant (p-values between 0.33 and 0.85).

Out of sample, the answer differs by horizon (Panel B). The VAR predicts the excess returns of ON-1W, ON-2W, and 1W-2W with R^2_{oos} of 38.7%, 25.4%, and 49.6%, and the term spread alone does about as well, so this is the classical predictability. For ON-1M no estimated model attains a positive R^2_{oos}, the spread alone attains 0.2%, and the SPA test does not reject. Out of sample, the EH fails only at horizons of one and two weeks.

Panel C shows what this predictability consists of. Without the quotes of the decision day, the VAR's R^2_{oos} falls from 39.1% to 0.3% for ON-1W and from 25.9% to 5.1% for ON-2W on the same dates, and from 46.7% to 26.2% for 1W-2W. A transitory error in today's long-rate quote enters both the predictors and, with the opposite sign, the excess return. Transaction-based rates support this reading: since 2018 the composite overnight rate changes on 89% of days, against about half of days for the transaction-based secured overnight financing rate and the broad and tri-party general collateral rates published by the Federal Reserve Bank of New York, and its daily changes correlate with theirs at only 0.40. A slowly moving premium, in contrast, explains little at these horizons: an exponentially weighted mean of past excess returns attains R^2_{oos} of at most 18.8%, and the best forecasts beat it for all three pairs (SPA $p \leq 0.015$).

The holdout from 21 April 2021, after the design was fixed, repeats this pattern for all four pairs. The VAR's R^2_{oos} is 62.0% for ON-1W and 41.1% for ON-2W, but only 14.1% and 9.4% without today's quotes. What survives out of sample is therefore statistically significant but small once quote timing is accounted for. The next question is whether a more flexible functional form would find more.

4.2. Does functional-form flexibility matter?

The flexible learners do not reveal departures that the linear model misses. The best of 19 learners falls short of the VAR for ON-1W and 1W-2W and exceeds it by 0.9 percentage points of R^2_{oos} for ON-2W (Table 1). Of 84 pairwise Diebold–Mariano comparisons between the VAR and the other 21 forecasts, none favours the alternative at the 5% level, and 40 favour the VAR (Supplement, Section S3). For ON-1W and ON-1M every learner is encompassed by the VAR after a Holm adjustment. For ON-2W, where 77% of the VAR's gain arises in windows that contain a month turn, five learners add information. The neural-network ensemble stays closest to the EH: its worst R^2_{oos} is -4.2% (ON-1M), against -17.3% for the VAR and -56.4% for random forests.

Agreement between the models is not proof of linearity. In simulations with known

departures (Supplement, Section S5), the linear models recover a linear signal that the flexible learners miss, and the flexible learners detect a nonlinear signal only when it explains about 30% or more of the variance, without turning it into accurate forecasts. The horse race therefore shows that relaxing the functional form does not change the inference, but it cannot exclude small nonlinear departures. For this reason we bound the economic value of any forecast, not only of the models we estimate.

[Insert Table 2 here]

4.3. Does the rejection matter economically?

Table 2 translates predictability into utility. Before costs, the VAR is worth at most 2.17 basis points a year (1W-2W) and 1.80 for ON-1W, and no forecast, even when the best is chosen ex post, is worth more than 2.17 on any pair. Costs then decide the sign. With DST's 4-basis-point spread, the VAR loses relative to the EH on every pair, and no forecast earns more than 0.83 basis points. With the observed spreads of each pair, between 1.5 and 4.0 basis points, the VAR earns 0.68 basis points for 1W-2W and loses on the other pairs, and no forecast earns more than 0.90. The VAR's break-even spread is 1.5 to 2.2 basis points. In the holdout, none of the seven forecasts evaluated there earns a positive fee after a 4-basis-point spread.

The last five columns bound the value of any forecast. On average, the realised excess return deviates from the EH forecast by only 3.5 to 6.0 basis points a year. This is the entire departure available to a forecaster, whatever its source, including quote noise. An investor with perfect foresight of its sign would earn between 3.4 and 4.1 basis points a year before costs and at most 2.3 after the observed spreads. Statistical predictability can thus be strong while economic failure is tightly bounded: whatever the forecasting model, departures at the short end are worth at most a few basis points a year.

5. Conclusion

At the short end of the US repo curve, the EH fails statistically but is bounded economically. A test that does not depend on a model of expectations confirms DST's economic verdict under weaker assumptions. In-sample rejections are uninformative; the out-of-sample failure is confined to horizons of one and two weeks, is captured by a linear function of current rates, and is largely tied to same-day quotes; and even perfect foresight would earn only a few basis points a year. These conclusions have limits. The bound is stated per unit of notional and for average bid–ask spreads, which widen in stress. The evidence rests on composite quotes for maturities up to one month, because the longer composites stopped updating in 2015. Transaction-based term rates and other currencies would test the bound further.

Declaration of competing interest

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Dennis Jung: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

The repo quotes are proprietary Bloomberg data and cannot be shared under the data licence. The transaction-based rates (secured overnight financing rate, broad and tri-party general collateral rates; Federal Reserve Bank of New York) and the federal funds target (FRED) are publicly available. The code that reproduces all results from these data is available from the author on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to support programming and text editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Table 1: **What Survives Out of Sample**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: In-sample test of DST				
Wald test, asymptotic p-value	< 0.001	< 0.001	< 0.001	< 0.001
Wald test, p-value simulated under the EH	0.535	0.845	0.325	0.840
Rejection rate of a true EH at 5% (%)	88	96	97	97
Panel B: Out-of-sample R^2 relative to the EH forecast (%)				
Spread only	34.6^{***}	25.8^{***}	0.2^{**}	47.2^{***}
VAR (DST information set)	38.7^{***}	25.4^{***}	-17.3	49.6^{***}
Ridge (regularised linear)	21.3^{***}	22.3^{***}	-18.0	48.2^{***}
Random forest (trees)	-54.4	-1.4^{***}	-56.4^{*}	27.3^{***}
Neural-network ensemble	1.9^{***}	3.4^{***}	-4.2	5.9^{***}
Best of 19 learners	21.7^{***}	26.3^{***}	-1.0^{*}	48.2^{***}
SPA p-value: no model beats the EH	0.011	0.000	1.000	0.010
Panel C: What the predictability consists of				
VAR without today's quotes, R^2_{oos} (%)	0.3^{***}	5.1^{***}	-21.1	26.2^{***}
Slowly moving premium (EWMA), R^2_{oos} (%)	0.7	8.7	-2.2	18.8
SPA p-value: no model beats the EWMA	0.015	0.004	0.884	0.002
Panel D: Holdout, 21 April 2021 to September 2026, R^2_{oos} (%)				
VAR	62.0^{***}	41.1^{***}	-47.2	12.0^{***}
VAR without today's quotes	14.1	9.4	-63.0	-2.5
Out-of-sample observations	4,666	3,277	3,088	3,332

The target is the annualised excess return of rolling the short repo over holding the long repo, Equation (1); its EH forecast is the prevailing mean. Panel A: Wald test of the VAR(5) restrictions of Della Corte et al. (2008) and its p-value and size in 200 samples simulated from a two-factor model in which the EH holds exactly (Supplement, Section S2). Panel B: R^2_{oos} of Campbell and Thompson (2008); stars: Clark and West (2007) test of $R^2_{oos} \leq 0$ at the 10%, 5%, and 1% level, Newey–West variance with at least as many lags as the horizon. All models use five daily lags of the two rates, the same rolling five-year windows, and the same tuning protocol (Table S1). The best learner is the maximum over 19 learners (regularised linear, kernel, tree-based, random-feature, and neural-network models); the SPA test of Hansen (2005) accounts for the search over all 22 forecasts. Panel C: the VAR with lags dated $t-1$ to $t-5$ instead of t to $t-4$, evaluated on the same dates; an exponentially weighted mean (EWMA) of past excess returns (half-life six months); and the SPA test with this mean as the benchmark. Panel D: forecasts made after the design was fixed on data to April 2021; '–': no clean quotes in the holdout.

Table 2: **Economic Value of Departures from the EH and Its Upper Bound**

	Spread	VAR				Best of 22 forecasts			Any forecast: upper bound				
	observed	Gross	Net 4 bp	Net obs.	Break-even	Gross	Net 4 bp	Net obs.	$\lvert xr - \overline{xr}\rvert$	Gross	Net 4 bp	Net obs.	Break-even
ON-1W	4.0	1.80	−1.41	−1.41	2.24	1.80	0.34	0.34	5.0	3.61	−0.11	−0.11	3.89
ON-2W	3.5	0.83	−1.45	−1.17	1.46	1.21	0.81	0.82	5.2	3.38	−0.05	0.37	3.94
ON-1M	3.5	0.33	−0.29	−0.21	2.12	1.11	0.83	0.85	6.0	3.57	1.00	1.32	5.55
1W-2W	1.5	2.17	−1.80	0.68	2.19	2.17	0.38	0.90	3.5	4.05	−0.71	2.26	3.41

Annualised performance fee $\mathcal{F}$ in basis points (bp) that an investor with quadratic utility and relative risk aversion of 5 would pay to switch from the EH strategy (prevailing-mean forecast) to a strategy based on the forecasts (Fleming et al., 2001; Della Corte et al., 2008). The position in the rolling strategy is $w_t = (\bar{\sigma}_t/\widehat{\sigma}_t)\,\mathrm{sign}(\widehat{\mu}_t)$, so the median position is about one unit of notional; a new tranche is opened every day and held for the life of the long repo. Net 4 bp: after a bid–ask spread of 4 basis points in rate terms on each change of a tranche, as in DST. Net obs.: after the observed spread of the pair, the average of the mean bid–ask spreads of its two repo maturities over the sample (column Spread, in basis points). Best: the largest fee of the 22 forecasts on each pair (chosen ex post). Break-even: the bid–ask spread at which the fee falls to zero (0 if it is not positive before costs). Upper bound: $\lvert xr - \overline{xr}\rvert$ is the average absolute deviation of the realised excess return from the EH forecast, in basis points a year, the total departure available to any forecast; Gross, Net, and Break-even refer to an investor with perfect foresight of the sign of the excess return in the same strategy. Out-of-sample forecasts from five lags of the two rates of each pair.

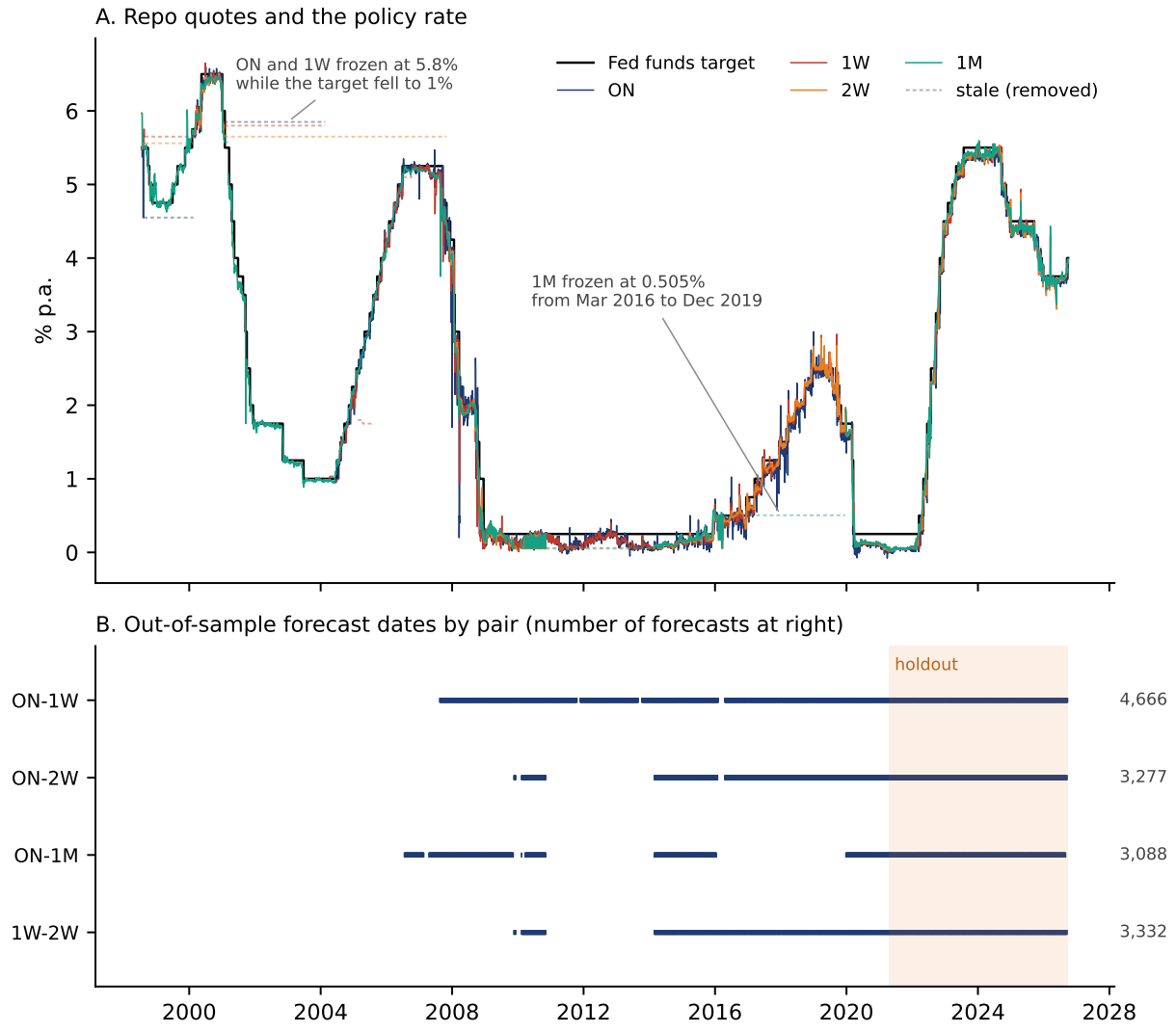


Figure 1: **Repo Quotes, Stale Quotes, and the Evaluation Samples.** Panel A: daily composite quotes of the overnight, one-week, two-week, and one-month general-collateral repo rates and the federal funds target (upper bound of the range from December 2008), in percent per annum, 23 July 1998 to 30 September 2026. Solid lines are the quotes used; faded dashed lines are stale quotes, runs of identical values longer than 20 business days, which are removed. Panel B: dates with an out-of-sample forecast of the excess return for each pair after the removal of stale quotes, with the number of forecasts at the right; the shaded area is the holdout from 21 April 2021, whose data were not used to design the study.